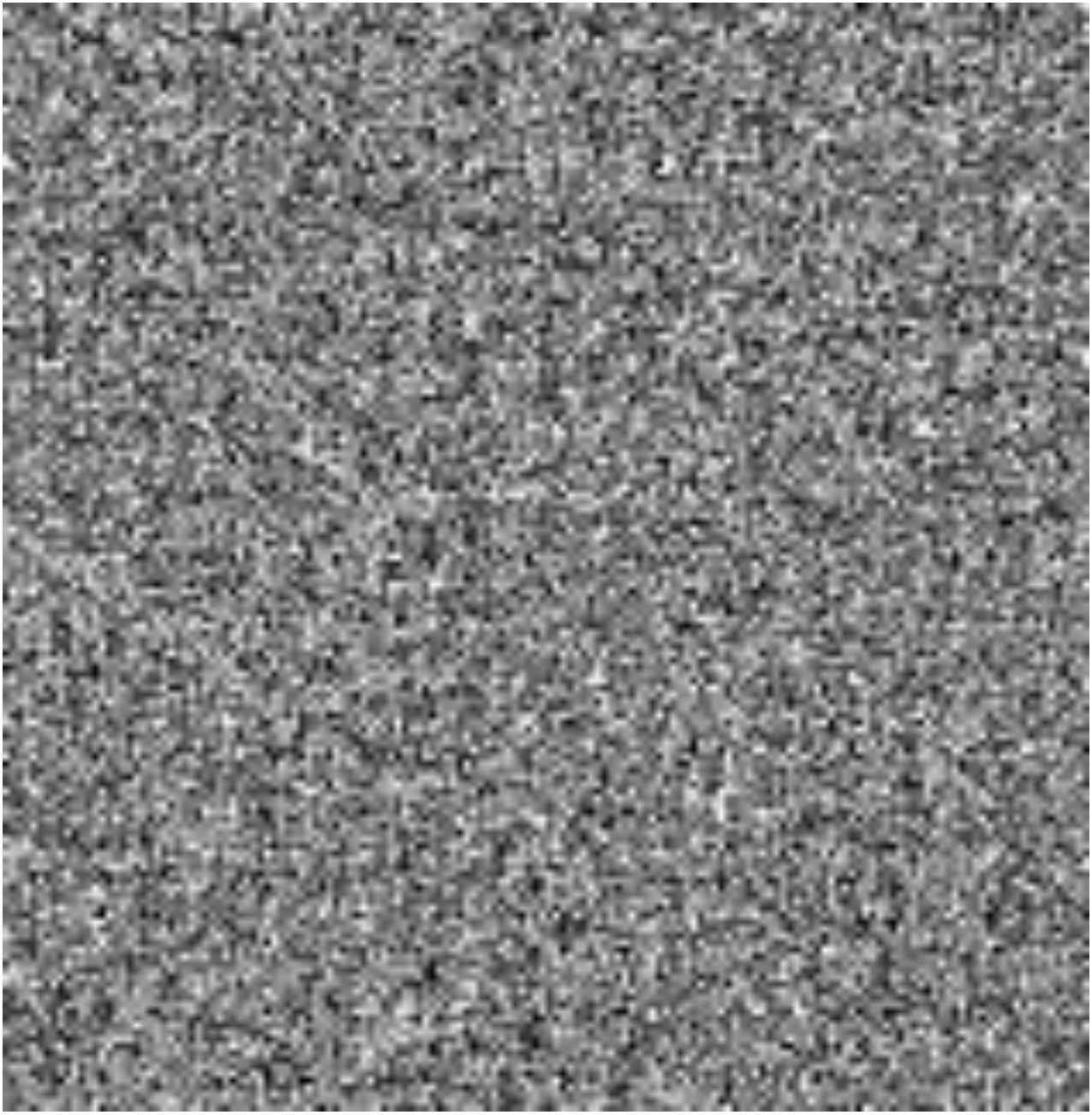


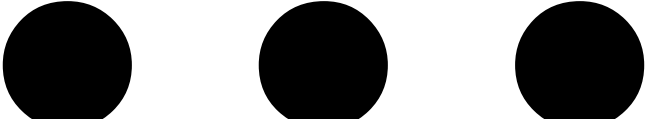
Noise

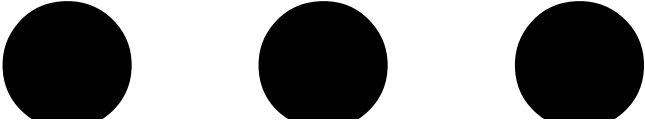
Output

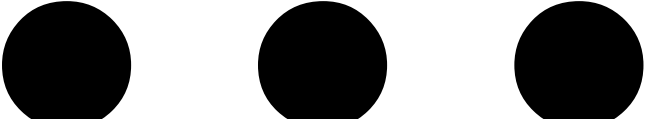
Denoising process

(Noise → Image)

T2I Generation







Noise

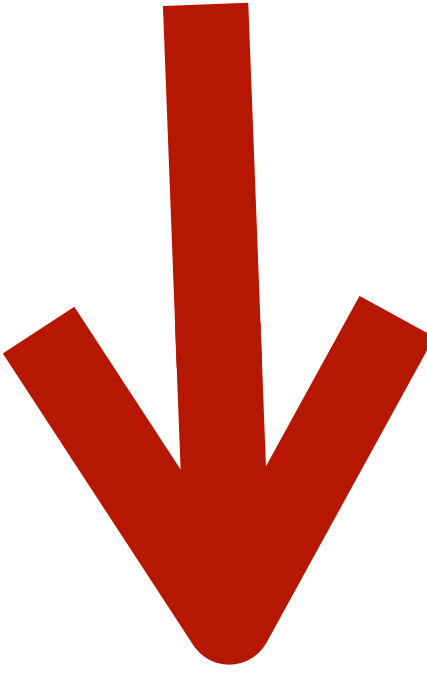
Output

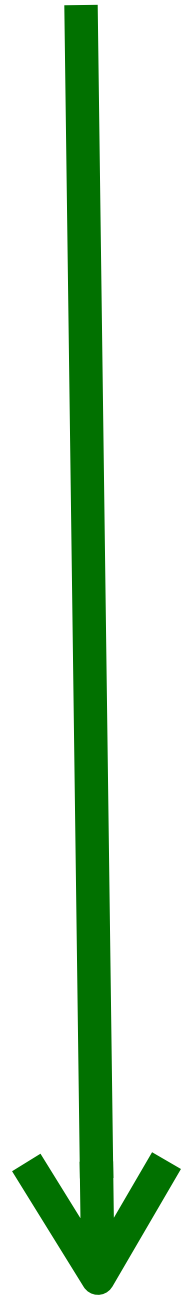
Denoising process

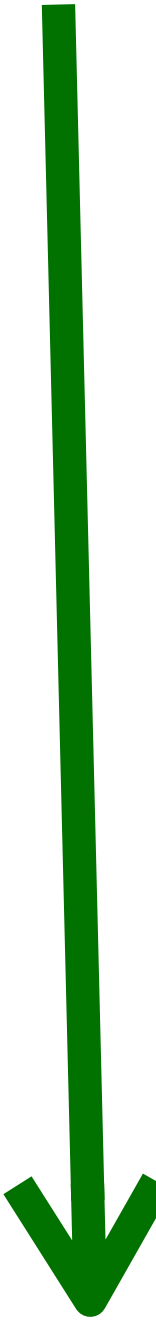
Cat

Training-free T2I Editing

(Image → Noise → Image)







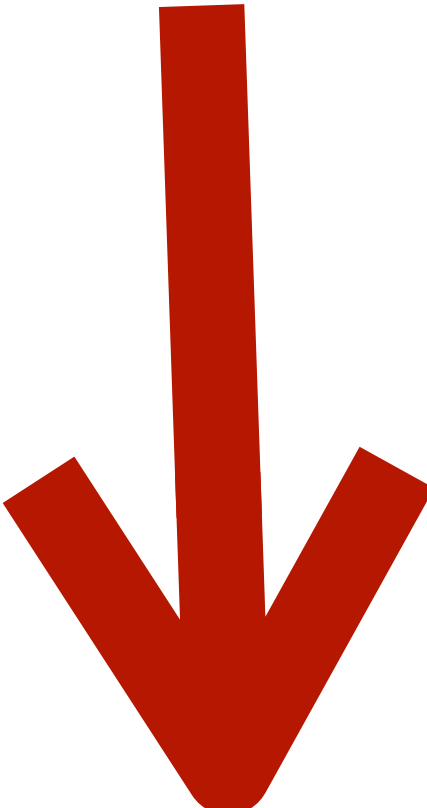
Tiger

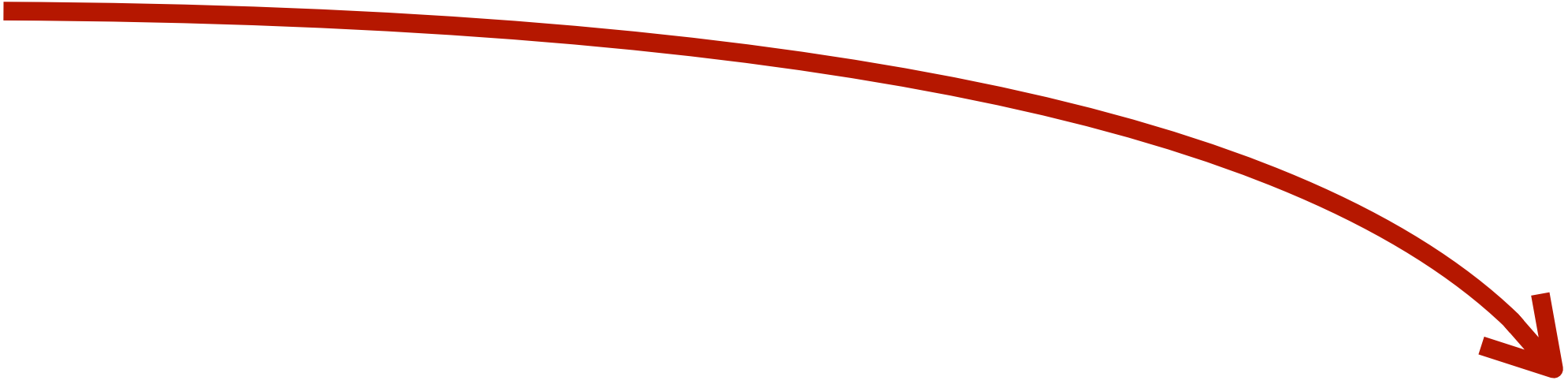
Noise

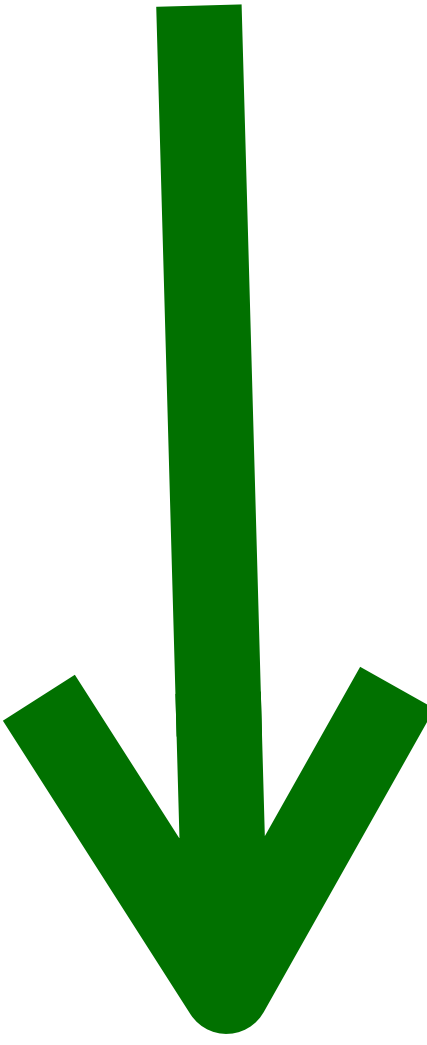
T2I Inversion

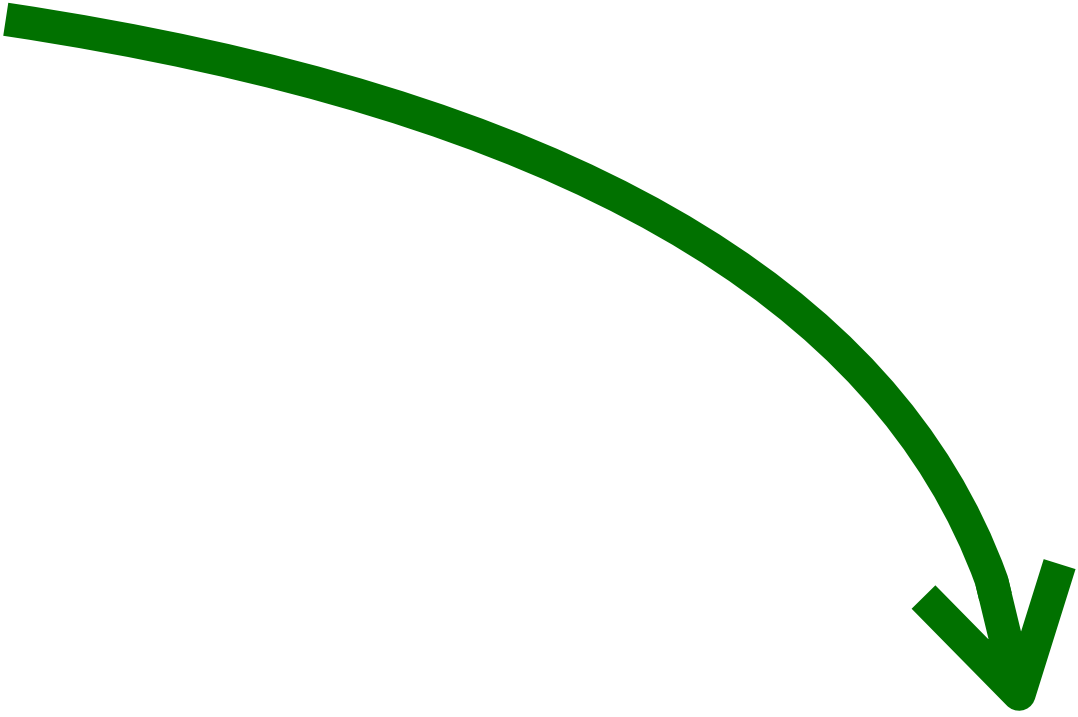
(Image → Noise)

Tiger

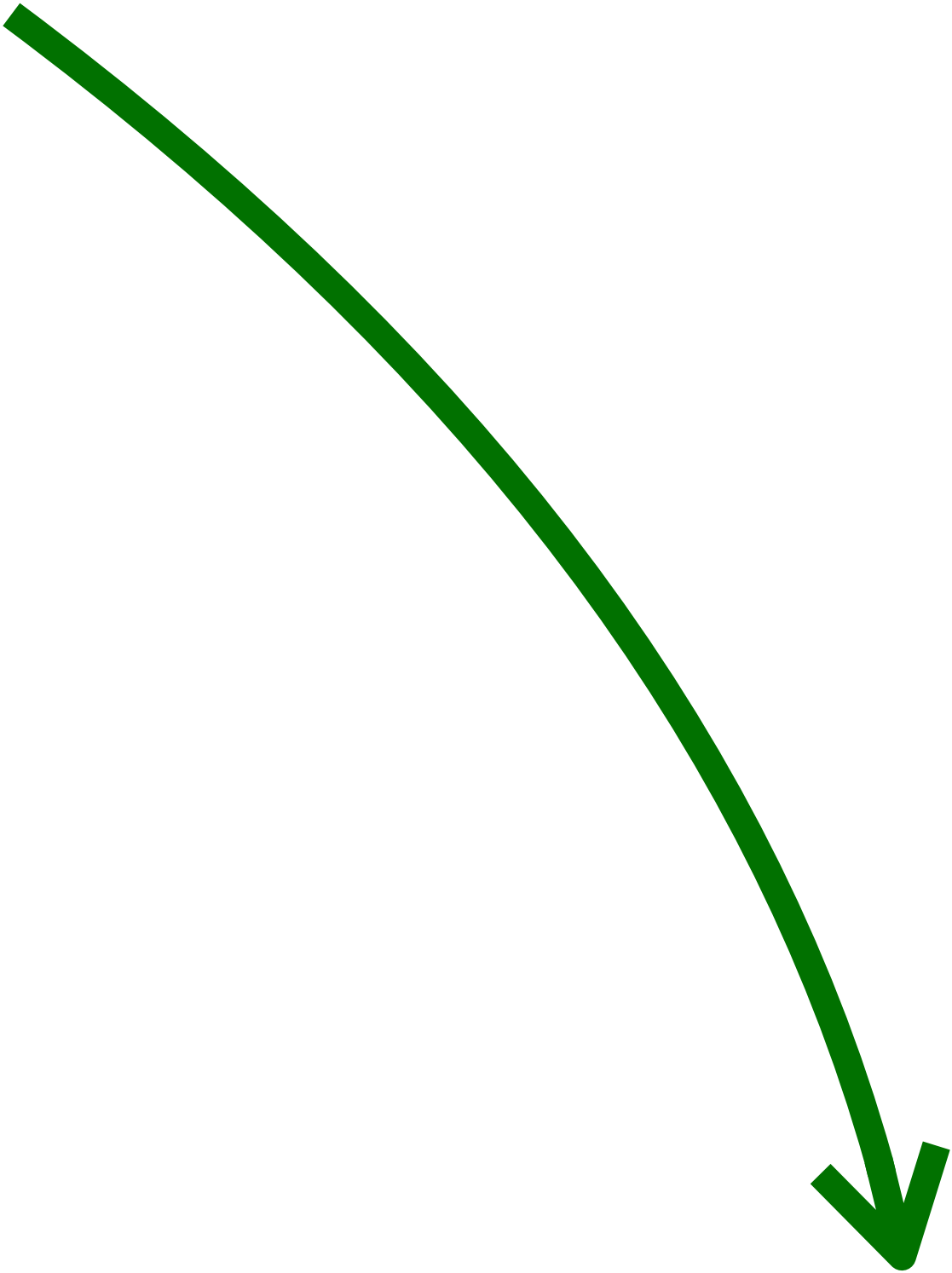


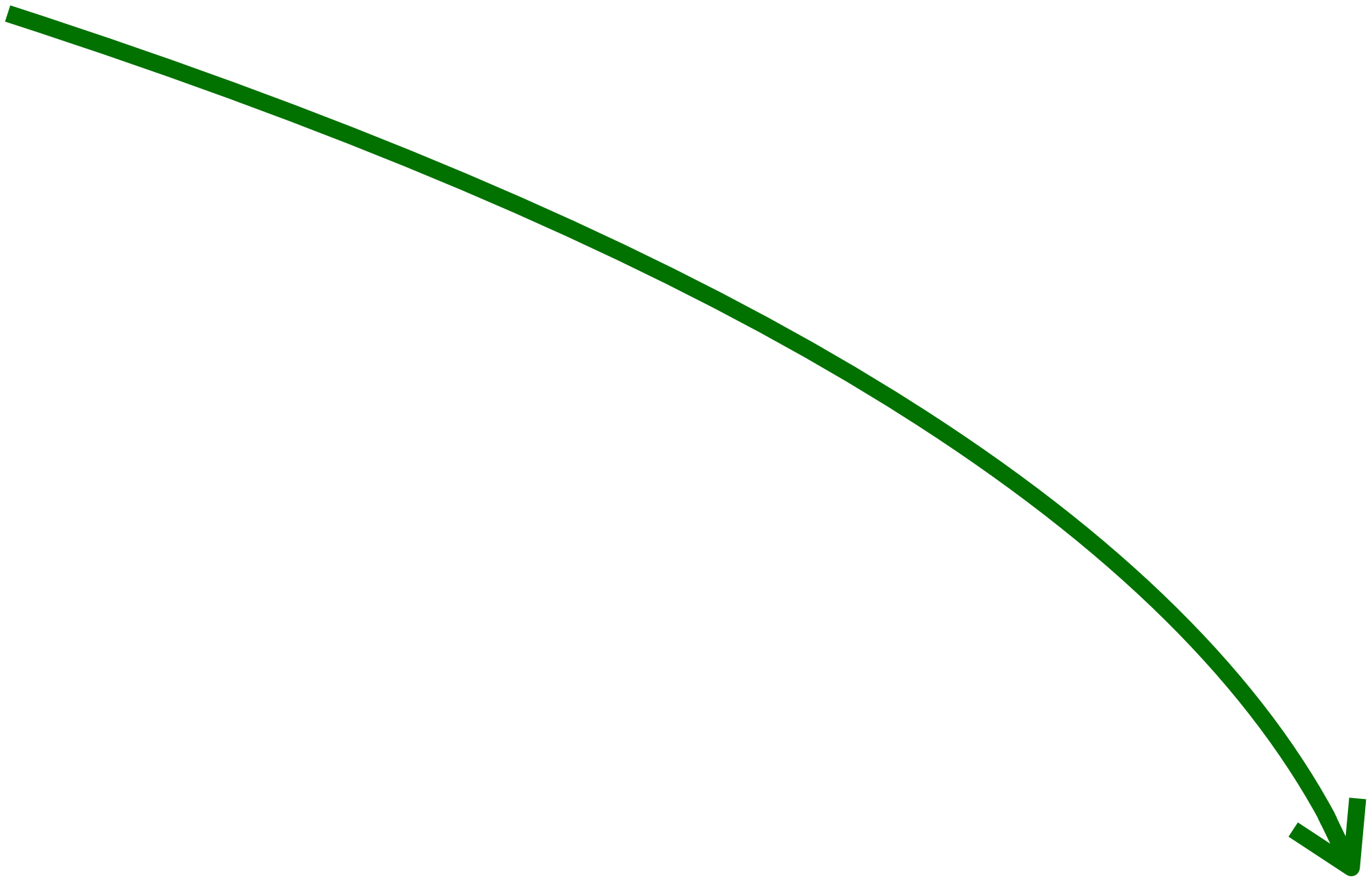






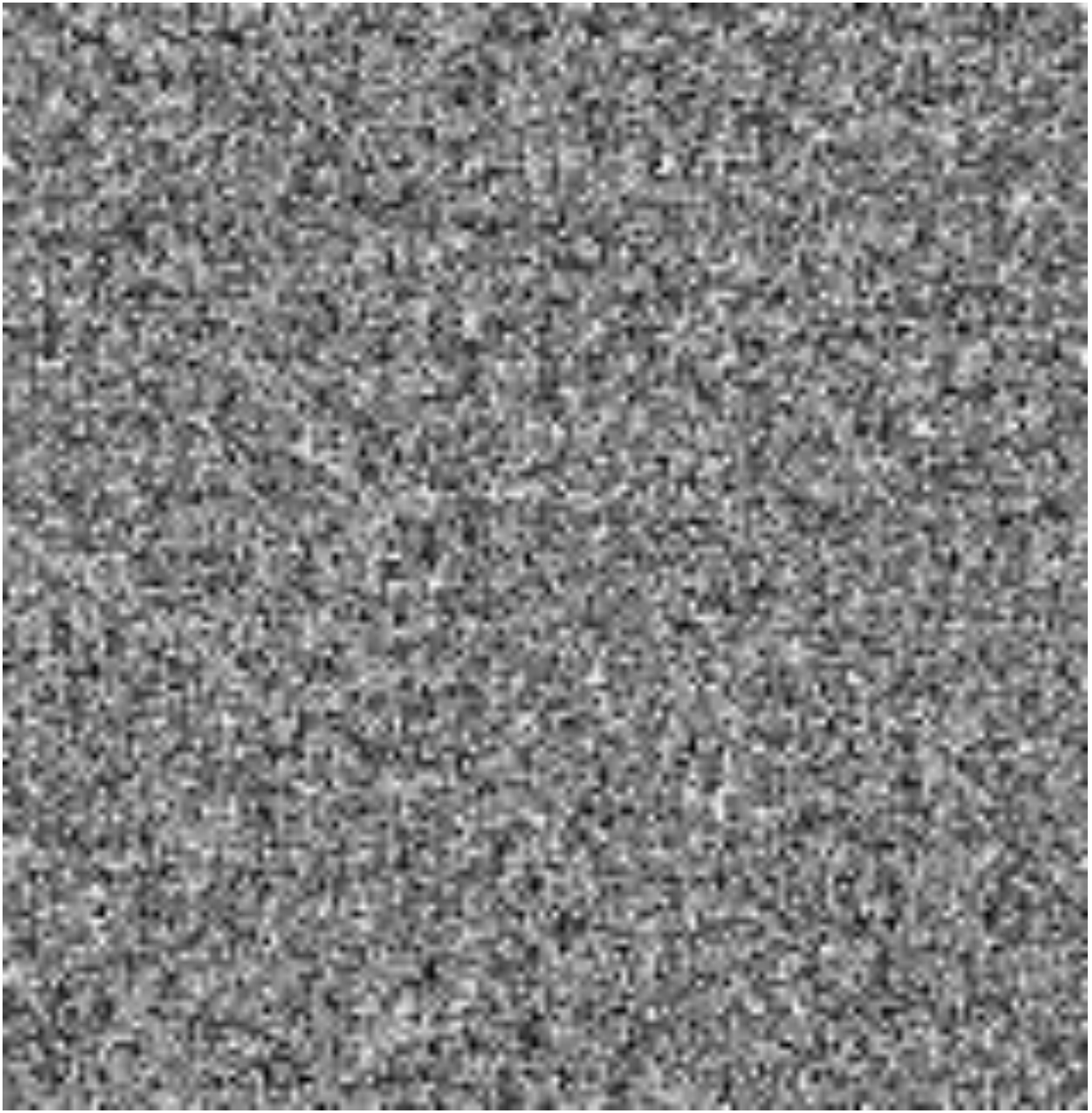
Text Guidance



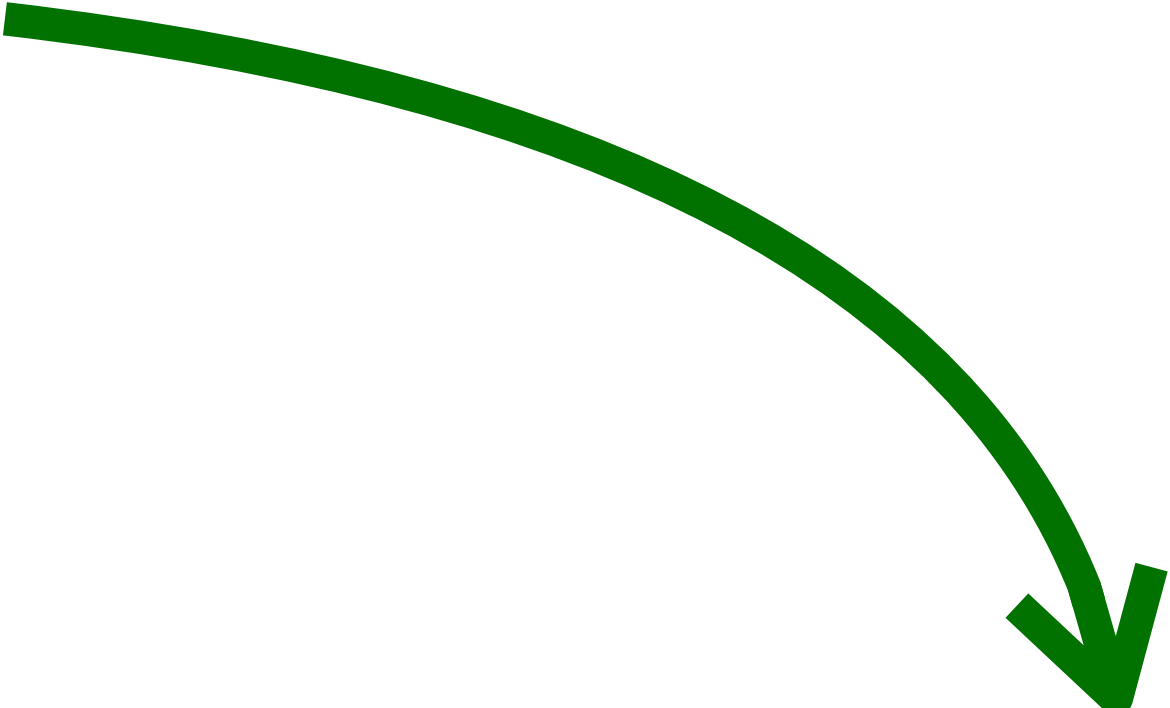


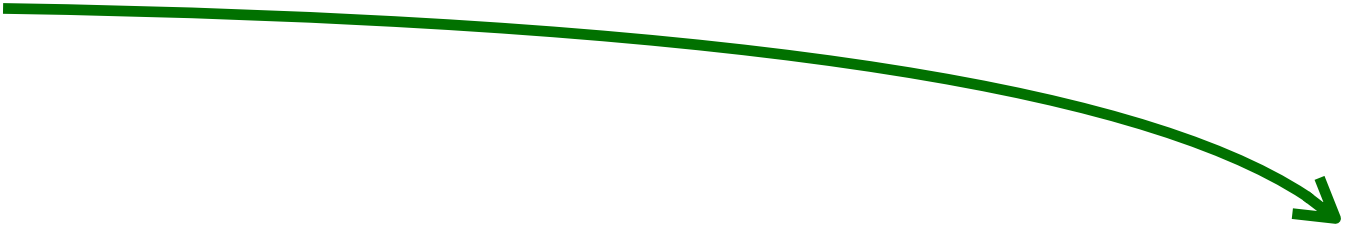
Goal #1

Fidelity / reconstruction

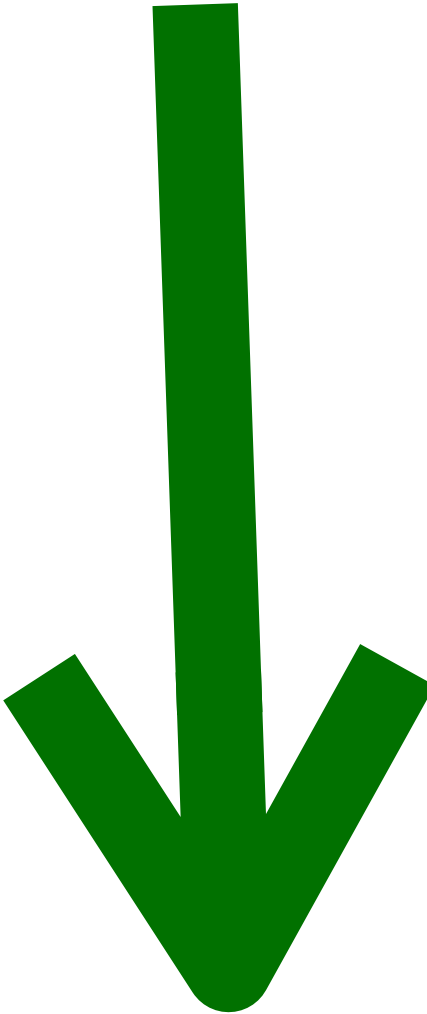


Tiger





Text Guidance



Instruction following

Goal #2:

Diffusion Models (DMs) & Rectified Flow Models (RFs)

Training-Free Editing Based on Generation Models

3

Inject reference info

Training-free T2I Editing

Multiple time-step denoising process

Fidelity / reconstruction

Goal #1:

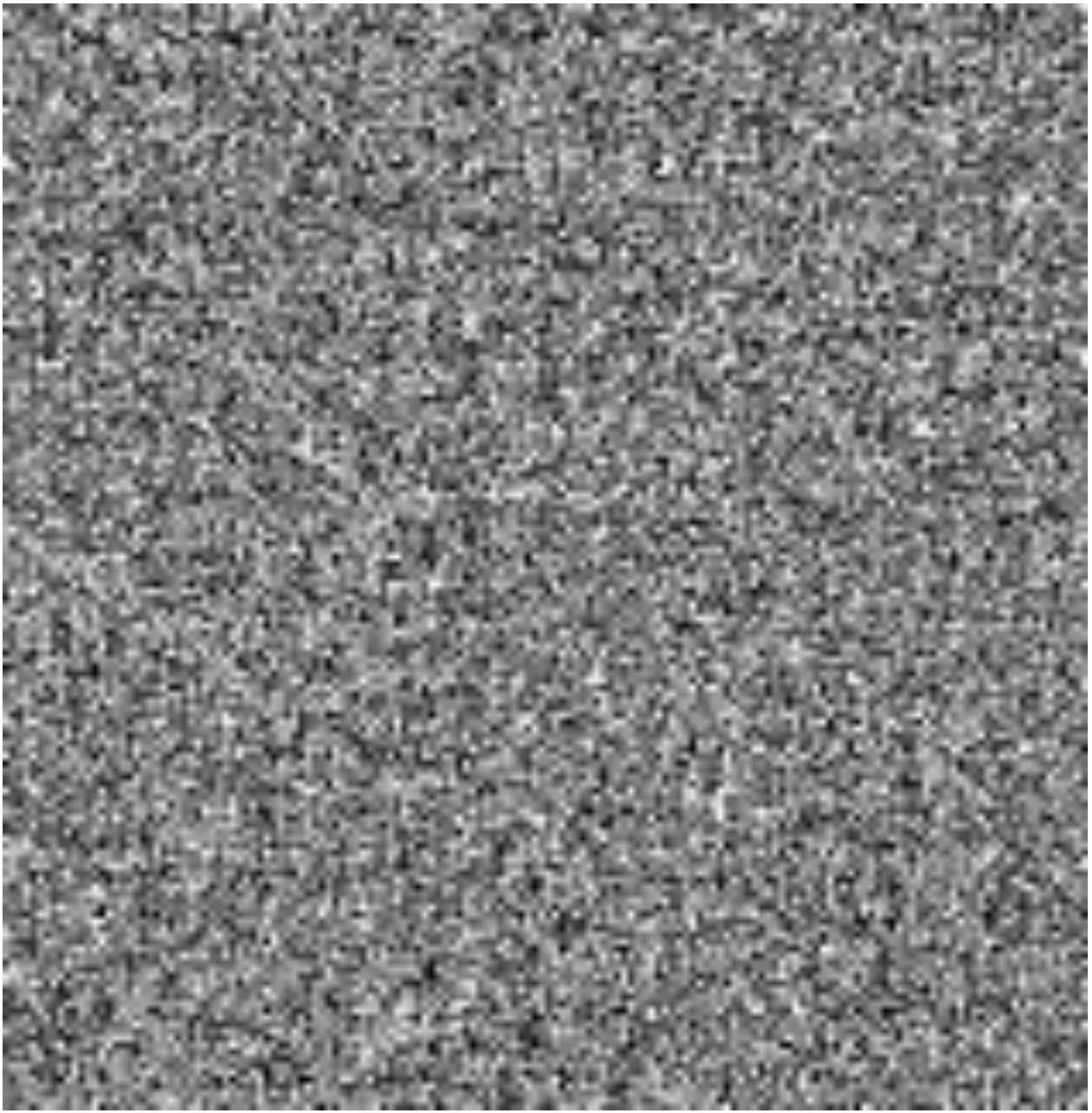
trajectory

Reference

Keep Visual Fidelity via Reconstruction

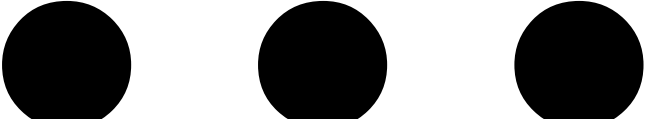
trajectory

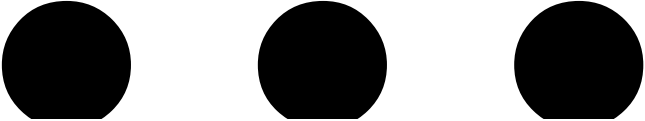
Editing

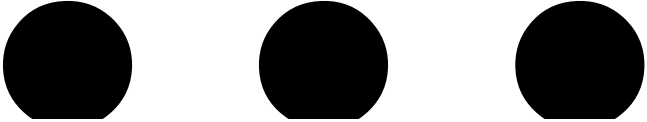


Noise

Output







Noise

Output

Cat

Tiger

Text Guidance

Instruction following

Goal #2: